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A meta-analysis of the correlation between self-regulated learning strategies and academic performance in online and blended learning environments

Yingying Zhao [a], Yixun Li [b], Shuai Ma [a], Zhihong Xu [a,*], Bingsheng Zhang [c]

[a] Texas A&M University, 400 Bizzell St, College Station, TX, USA
[b] The Education University of Hong Kong, 10 Lo Ping Road, Tai Po, Hong Kong
[c] StataCorp, 4905 Lakeway Dr, College Station, TX, USA

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ABSTRACT

The present meta-analysis examines the association between different self-regulated learning (SRL) strategies and academic performance in online and blended learning environments. A database search identified a total of 42 studies from 37 articles from 2011 to 2022, containing 115 effect sizes from 11,014 participants in K-12 and higher education. The overall effect size between SRL strategies and academic performance was .14. We did not detect any significant moderation effects except for educational level. To examine the differences in how SRL strategies are linked to academic performance, we compared the relationships between different SRL strategies with academic performance at two different levels: 16 specific SRL strategies (operational level) and four categories of SRL strategies (conceptual level). We found that, at the conceptual level, all four SRL strategies were significantly correlated with academic performance. On the operational level of SRL strategies, the results showed that rehearsal, organization, metacognitive strategy, time management, effort regulation, and general SRL strategies were effective strategies associated with academic performance. The present findings contribute to the future advancement of SRL research in online learning and guide the development of targeted SRL interventions.

1. Introduction

As online and blended instructional models continue to proliferate across educational settings, learners are gaining enriched and flexible learning experiences while facing new challenges (Hwang et al., 2021; Mason et al., 2013). Online learning refers to the use of the Internet to obtain learning materials and support, as well as to interact with instructors and peers within a virtual learning environment (e.g., Ally, 2008). Blended learning integrates traditional face-to-face instruction with online learning components (Garrison & Kanuka, 2004). Compared with traditional face-to-face learning, effective online and blended learning demands a higher degree of self-direction from students. Online learning environments may vary in terms of teacher presence and interactivity, ranging from asynchronous courses, where learners independently navigate materials, to synchronous formats that allow real-time interactions

* Corresponding author.
E-mail addresses: fhlzhaoyy@tamu.edu (Y. Zhao), lyixun@eduhk.hk (Y. Li), shuai_ma2021@tamu.edu (S. Ma), xuzhihong@tamu.edu (Z. Xu), shanby84@gmail.com (B. Zhang).

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between students and instructors. In these contexts, students need to set goals, control their own learning agenda, and actively adapt their learning strategies to maintain engagement (e.g., Murphy & Cifuentes, 2001; Pérez-Álvarez et al., 2018; Xu et al., 2023). These practices can be conceptualized as self-regulated learning (SRL; Azevedo, 2008), an important predictor of learners' academic performance (e.g., Fernandez-Rio et al., 2017; León et al., 2015; Xu et al., 2022; Xu et al., 2023). To support students' online learning with scientific research, the present meta-analysis aims to understand the relationship between SRL strategies and academic performance in online and blended environments across a wide age range of students from kindergarten to university.

Several meta-analyses have documented the association between SRL strategies (i.e., strategies used in the SRL process) and academic performance, despite the fact that they either did not consider online learning elements (e.g., Dent & Koenka, 2016; Jansen et al., 2019; Li et al., 2018; Richardson et al., 2012), or focused on online learning (e.g., Broadbent & Poon, 2015). Ample evidence suggested that students' use of SRL strategies could be improved by interventions (e.g., Dignath et al., 2008; Jansen et al., 2019; Xu et al., 2022; Zheng, 2016). In practice, students may apply a variety of SRL strategies to achieve their learning goals (Zimmerman, 1990). Note that different SRL strategies are not equally effective: in prior empirical studies, some SRL strategies, such as metacognition, appear to produce larger effect sizes on academic performance than others, such as rehearsal (e.g., see Richardson et al., 2012, for a meta-analysis). To guide the future development of targeted SRL interventions in online and blended learning environments, a critical step is to identify the SRL strategies that are most strongly linked to students' academic achievement through a comprehensive meta-analysis, which is the gap that this work sought to fill.

1.1. Self-regulated learning (SRL) and SRL strategies

Self-regulated learning (SRL) is an active and constructive learning process whereby students set goals and attempt to regulate their cognition, motivation, and behavior (Zimmerman, 1986). Since this broad definition, SRL theories and empirical studies about SRL have emerged in recent decades. Five widely acknowledged SRL models have been proposed by Boekaerts (Boekaerts & Niemivirta, 2000), Borkowski (1996), Pintrich (2000), Winne (Winne & Hadwin, 1998), and Zimmerman (2000) (Puustinen & Pulkkinen, 2001). A more recent review by Panadero (2017) revisited different SRL models and concluded that the models proposed by Zimmerman (2000) and Pintrinch (2000) were more comprehensive and have broader utilization compared to other classical models such as Boekaerts' (Boekaerts & Niemivirta, 2000); one key reason is that both models include relatively complete versions of specific SRL sub-processes.

There is disagreement among SRL models regarding the role of motivation (Jansen et al., 2019; Panadero, 2017). Some researchers include motivation as an integral part of SRL (e.g., Boekaerts, 2011; Zimmerman, 2000). Others differentiated motivation/affect from the regulation of motivation/affect (e.g., Efklides, 2011; Ning & Downing, 2012; Pintrich, 1999). For example, Pintrich (1999) highlighted three different motivational beliefs - task value, goal orientation, and self-efficacy - and how they can promote and sustain SRL. The Motivated Strategies for Learning Questionnaire (MSLQ; Pintrich et al., 1991) consists of two separate subscales, namely motivation scales of three parts (value, expectancy, and affective components) and learning strategies (cognitive, metacognitive, and resource management strategies).

While we acknowledge the significant role of motivation in self-regulated learning, the main focus of the current review is on the SRL strategies that students use in the learning process. Consistent with previous review papers (e.g., Broadbent & Poon, 2015; Dent & Koenka, 2015; Jansen et al., 2019), we decided not to include motivation from the primary focus of this research to maintain comparability with these earlier reviews. This allows us to concentrate on the strategies that are directly involved in the learning process, providing a clear and focused analysis of SRL strategies.

In the SRL process, students may choose a variety of strategies to achieve their learning goals (Zimmerman, 1990). Empirical evidence suggests that higher-achieving students tend to use SRL strategies more effectively and frequently than lower-achievers (Zimmerman, 2002). Pintrich's (2004) conceptual framework for SRL strategies is one of the most influential models in the field, which conceptualizes three major types of SRL strategies among students: *metacognitive strategies* (i.e., students' awareness and control of cognition), *cognitive strategies* (i.e., the basic and complex approaches that student use for processing information while performing learning tasks), and *resource management strategies* (i.e., time and environment management strategies that help students regulate external resources).

Pintrich et al.'s (1991) MSLQ has been the most widely used tool to measure students' use of SRL strategies (Panadero, 2017). Across numerous studies conducting the MSLQ, at least 12 specific types of SRL strategies could be identified: one type of *metacognitive strategy* (i.e., the awareness and control of cognition including processes such as planning, monitoring, and regulating); five types of cognitive-related strategies (i.e., students' use of *rehearsal* [e.g., repeat words to oneself], *elaboration* [e.g., paraphrasing], *organization* [e.g., using flowcharts], *critical thinking* [e.g., evaluating ideas critically], and *general cognitive strategies* [appeared in some studies due to the lack of specifications, such as Kim et al., 2014]); five types of resource-management-related strategies (i.e., *time and environment management* [e.g., managing time well and finding appropriate study places], *effort regulation* [e.g., keeping studying hard while facing difficult], *peer learning* [e.g., using learning groups], *help-seeking* [e.g., asking for help from others], and *general resource management strategies* [appeared in some studies without specifications, such as Templeman, 2020]), in addition to the *general SRL strategies* (i.e., when researchers did not specify the SRL strategies in their work, such as in Oladejom, 2012). All these SRL strategies were specific enough in operationalization at the practical level.

1.2. SRL strategies and academic performance in previous meta-analyses

Numerous meta-analyses have examined the relationships between students' SRL strategies and academic performance, both in traditional face-to-face environments (e.g., Dent & Koenka, 2016; Jansen et al., 2019; Li et al., 2018; Richardson et al., 2012) and in online learning contexts (e.g., Broadbent & Poon, 2015). Jansen et al. (2019) sought to examine the mediation of SRL strategies (activities) in the effect of SRL intervention on academic achievement in higher education; they did not compare different SRL strategies, but part of their results found a significant correlation between SRL strategies and academic achievement, with an overall correlation of $r = .25$ across 70 articles. Dent and Koenka (2016) focused on the relationship between SRL and academic performance (achievement) across children and adolescents, particularly on comparisons among the metacognitive strategies (the overall correlation $r = .20$ across 61 articles) and that among cognitive strategies (the overall correlation $r = .11$ across 57 articles).

Richardson (2012) analyzed 242 datasets from 217 articles published between 1997 and 2010 that examined the relationships between university students' psychological constructs, including SRL strategies, and their academic performance (i.e., Grade Point Average; GPA). While without calculating the overall correlation, they identified the 10 specific SRL strategies covered in the MSLQ (expect for the *general SRL strategies* explained in the previous section) in addition to *concentration* (i.e., the ability to direct and maintain attention during academic tasks; Weinstein et al., 1988), with eight of them significantly related to GPA: *metacognition* ($r = .18$), *time management* ($r = .22$), *effort regulation* ($r = .32$), *peer learning* ($r = .13$), *elaboration* ($r = .18$), *critical thinking* ($r = .15$), *help seeking* ($r = .15$), and *concentration* ($r = .16$). Moreover, Li et al. (2018) focused on Chinese students' SRL and academic performance across 59 studies published from 1998 to 2016; the overall effect size (Cohen's d) calculated in this meta-analysis was .435, and the effect sizes of *cognitive strategies* (task strategies; $d = .600$) and *critical thinking* (self-evaluation; $d = .717$) were relatively higher than other strategies such as *metacognitive strategies* (metacognitive monitoring; $d = .388$).

The only published correlational meta-analysis pertaining to online learning, as per our investigation, is the work of Broadbent and Poon (2015), which synthesized data from 12 studies sourced from 11 articles, with a specific emphasis on adult learners. They identified nine SRL strategies out of the 11 specific SRL strategies covered in the MSLQ (except for the *general SRL strategies*). The weighted mean correlation across all effect sizes was significant ($r = .13$); five SRL strategies were significantly associated with academic performance, *metacognition* ($r = .06$), *time management* ($r = .14$), *effort regulation* ($r = .11$), *critical thinking* ($r = .07$), and *help seeking* ($r = .09$), whereas the other four SRL strategies were not, *peer learning* ($r = .30$), *elaboration* ($r = .00$), *rehearsal* ($r = -.03$), and *organization* ($r = .00$). Note that *peer learning* and *elaboration* were found to be significantly related to GPA in Richardson's (2012) meta-analysis in face-to-face environments, shedding light on the potentially different association between SRL strategies in and out of online learning. While Broadbent and Poon's study provided valuable insights into SRL strategies in online learning contexts, our study extends their work by including a broader range of grade levels rather than focusing on college learners. Additionally, we incorporate a larger sample of studies, offering a more comprehensive of SRL strategies across diverse educational contexts.

1.3. SRL strategies and academic performance in online and blended learning environments

The importance of SRL in online and blended learning environments has received more attention in recent years (e.g., Broadbent & Poon, 2015; Wong et al., 2021; Xu et al., 2022; Xu et al., 2023), as students' success in online learning heavily depends on their autonomy and the ability to actively apply SRL strategies (Pérez-Álvarez et al., 2018). Online learning has the potential to provide learners with access to diverse learning resources and opportunities for collaborative learning that may not be as readily available in offline settings (Santiago et al., 2021). Asynchronous online learning also allows students to study at their own pace and place. It is important to point out that effective learning in online and blended environments requires learners' use of SRL strategies (Azevedo, 2008); students must control their own learning agenda and take the initiative to learn content and interact with teachers and peers (Murphy & Cifuentes, 2001). In this regard, it becomes imperative to scrutinize and discern the relationships between SRL strategies and academic performance in online and blended learning environments. This undertaking forms the core objective of the present meta-analysis, underscoring its significance in advancing our understanding of learning outcomes in digital and hybrid educational settings.

1.4. The present study

To expand our knowledge of SRL strategies in online environments, the first goal of this work was to compare different SRL strategies on their connections to students' academic performance, particularly in empirical studies conducted in online and blended contexts. As students' educational level may have an impact on SRL effects (e.g., Panadero, 2017), unlike previous meta-analyses which heavily concentrated either on children and adolescents (e.g., Dent & Koenka, 2016; Li et al., 2018) or college students (e.g., Broadbent & Poon, 2015; Jansen et al., 2019; Richardson et al., 2012), we included students at all ages for a comprehensive picture.

Pintrich et al.'s (1991) MSLQ was used as a reference for defining and categorizing SRL strategies, as it provides a comprehensive list of SRL strategies that covered categorization of SRL activities in previous meta-analyses (e.g., Broadbent & Poon, 2015; Jansen et al., 2019). For conceptualization, our second focus was to conduct a higher-level theoretical comparison of various SRL strategies. To achieve this, we grouped these strategies into categories that align with Pintrich's (1999; 2004) theoretical framework, which includes *metacognitive*, *cognitive*, and *resource management* strategies, in addition to *general SRL strategies*, i.e., when the included studies did not specify the SRL strategies. This approach is one of the earliest empirical tests of Pintrich's (2004) SRL model using a meta-analytic methodology.

Lastly, for a careful design, five theoretical and methodological moderators were examined to determine whether they influenced the relationships between SRL strategies and academic achievement. Specifically, we investigated six moderators based on previous reviews and empirical research: article type (published journal articles vs. unpublished doctoral dissertations for detecting publication bias), learning context (online vs. blended, as the frequency of SRL strategies use and the effect sizes on grades varied across learning contexts, e.g., Broadbent, 2017), subject type (STEM vs. non-STEM, because SRL strategies were found more predictive of academic achievement among STEM disciplines than that of non-STEM ones, e.g., Li et al., 2018), educational level (K-12 vs. undergraduate vs. graduate, as the relation between SRL and academic achievement was found to be varied across grades, e.g., Dent & Koenka, 2016), measures of SRL strategies (questionnaires vs. behavioral coding), and measurements of academic performance (existing scores in schools vs. other instruments used in research) for the potential influence of measurement characteristics on the relationships (e.g., Jansen et al., 2019).

To sum up, the present study aimed to examine the association between students' use of SRL strategies and their academic performance in online and blended learning environments via the below four research questions.

RQ1. What is the overall correlation of SRL strategies on academic performance in online and blended learning environments?

RQ2. How do the study characteristics (e.g., learning contexts, educational level, academic measure) moderate the relationships between SRL strategies and academic performance in online and blended learning environments?

RQ3. What is the association of each specific SRL strategy (as identified in the included studies) on academic performance in online and blended learning environments?

RQ4. What is the association of each major type of SRL strategies (metacognitive, cognitive, resource management, and general SRL strategies) on academic performance in online and blended learning environments?

2. Method

2.1. Inclusion criteria

To systematically identify relevant studies for inclusion in the current meta-analysis, a set of inclusion criteria was established based on our research objectives and scope. Studies were included if they (1) were written in English; (2) were either published peer-reviewed journal articles or unpublished doctoral dissertations; Conference proceedings were excluded as they often lack a rigorous peer-review process and typically provide insufficient details on methodology and results. (3) were available online from 2011 to 2022, when the search was conducted. We chose 2011 as the starting year when the *Handbook of Self-Regulation of Learning and Performance* was published (Zimmerman & Schunk, 2011), consolidating theoretical and empirical advancements in SRL and providing a foundation for future research. Moreover, there has been significant progress in educational technology through the integration of digital tools and online learning platforms like MOOCs, which emerged around 2011 (Palacios Hidalgo et al., 2020). Granić's (2022) systematic literature review on the acceptance and adoption of educational technology from 2003 to 2021 also indicated that most studies have been published in the last decade. (4) reported at least one correlation coefficient between students' SRL and their academic achievement, as defined in the papers by the author(s); (5) included a component of online learning (either online or blended learning environment). We did not need to set up any rules for the age range of the participants, as participants of all ages were considered in this work.

The main focus here was on how different SRL strategies used by students were related to their academic achievement. Therefore, particular attention was paid to the specific SRL strategies addressed in the included studies. We expected to find at least nine SRL strategies (1, *metacognition*; 2, *time management*; 3, *effort regulation*; 4, *peer learning*; 5, *elaboration*; 6, *rehearsal*; 7, *organization*; 8, *critical thinking*; and 9, *help seeking*) based on a recent relevant meta-analysis (Broadbent & Poon, 2015). In the included articles, time management and environment management were both considered to be time management. As such, we stick with time management in the following. Given that the present scope was broader than Broadbent and Poon's (2015), we anticipated that additional SRL strategies would be identified in the included studies.

We also sought to clarify if specific SRL strategies were informative in themselves, or if it would be meaningful to group them into fewer, more conceptually based categories. Based on the existing key theoretical framework in SRL (Pintrich et al., 1991), we expected to find four broad types of SRL strategies in the literature: cognitive strategies (e.g., *rehearsal*, *elaboration*, *organization*, *critical thinking*, and *general cognitive strategies*[1]), metacognitive strategies (e.g., *planning* and *monitoring*; consistent with the definitions of metacognition processes in MSLQ), resource management strategies (e.g., *time management*, *effort regulation*, *peer learning*, and *help seeking*), and general SRL strategies (when a SRL strategy was not specified by researchers in the included studies), following the categorization in the MSLQ.

2.2. Searches

Guided by the above inclusion criteria, specific search terms were developed based on the key literature, grouped by the three key concepts of focus: self-regulation, academic achievement, and online and blended learning environments. For self-regulation, key search terms were self-regulation, cognitive strategy, metacognitive technique, rehearsal, elaboration, monitoring, self-evaluation, etc. For academic achievement, key search terms were achievement, academic ability, grade, test, indicator, academic performance, exam, GPA, etc. For online and blended learning environments, the key search terms were online, internet, virtual, web-based, e-learning, distance education, etc. Our search was broad, intending to cover all forms of online learning, which was a component of blended

[1] Cognitive strategies that were unspecified in the articles and thus cannot be categorized into the proceeding four specific cognitive strategies.

learning. In the screening process, we used blended or hybrid learning as one of the inclusion criteria to ensure we included studies conducted in blended or hybrid learning environments. In addition to the adjustment of subject terms in the databases, synonyms were developed for each concept for searches in the abstract and title fields. With these search terms, a systematic search was conducted in the five major databases in educational and psychological research: ERIC (EBSCO), APA PsycInfo (EBSCO), Education Source (EBSCO), Academic Search Ultimate (EBSCO), and Computer Source (EBSCO). A complete list of subject terms used in the five databases was uploaded to the Open Science Framework (https://osf.io/48r69/). The combined searches yielded a total of 7171 records. After removing the duplicates, 5458 references were obtained.

First, two authors discussed the inclusion and exclusion criteria in-depth, and both reviewed the first half of the articles. When there was a dispute, the two authors discussed and reached an agreement. Then, the two authors screened the rest of the articles separately, following the above inclusion and exclusion criteria. The first round of screening was based on the titles and abstracts of the articles, which reached 103 records. The same two authors conducted the full-text screening, achieving an inter-rater reliability of 93%. Discrepancies were resolved through discussion with a third author. After the second round of screening, 66 articles and dissertations were excluded for not meeting the 4th criterion (absence of reported correlation coefficients between students' SRL and their academic performance) or the 5th criterion (lack of focus on an online learning environment), as outlined above. The final pool consisted of 37 articles encompassing 42 independent studies, with five of them including two studies each. Fig. 1 illustrates the screening process in a PRISMA flow chart.

2.3. Coding

A codebook was designed to guide the coding of the key information and correlation coefficients directly from each included study based on the report and definitions written in the articles. All included articles were then coded following the codebook closely, including authors, publication year, article type (e.g., published journal article), number of participants, age and/or grade of participants (e.g., undergraduate), the SRL strategy used by students (e.g., rehearsal), the type of SRL measure (e.g., questionnaire), the type of academic performance measure (e.g., existing scores in a school course or researcher-designed assessment), the subject type studied (e.g., science), and the respective correlation coefficient (i.e., Pearson's r) representing the relationship between students' use of a SRL strategy and their academic performance in a particular measure. Each correlation coefficient was coded individually. As such, there could have been multiple coefficients from one single study if it examined more than one SRL strategy or included more than one task of academic performance or both. In the cases in which Pearson's r was not reported in the articles, we emailed the corresponding author(s) and asked for their help with the data to support our calculations of effect sizes. Two authors coded each study separately and

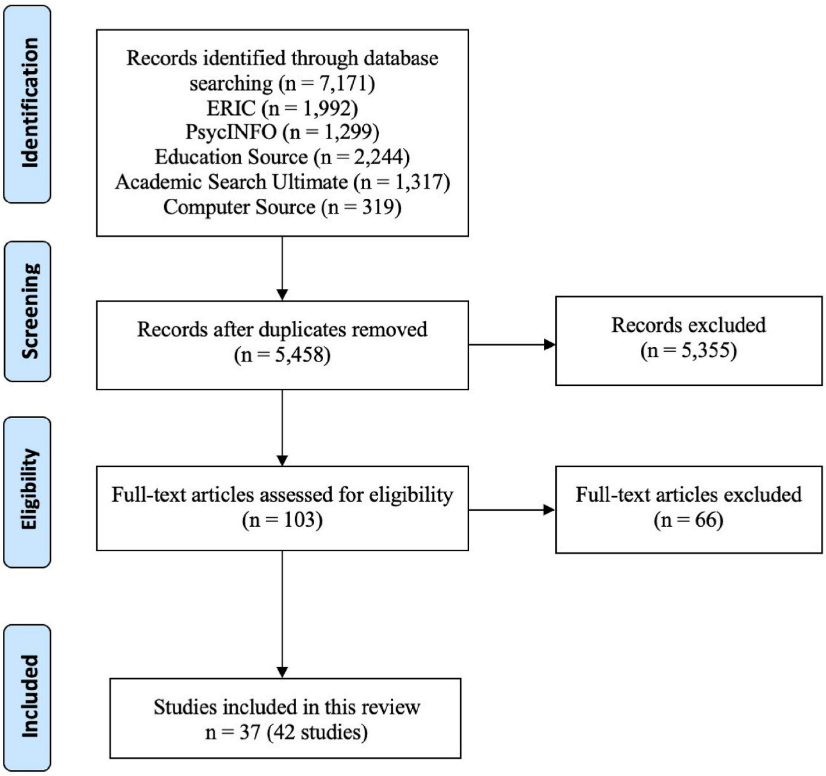

Fig. 1. Flow chart of search and identification procedures.

reached high inter-rater reliability (Kappa = .91). When there was a discrepancy in coding, another expert was consulted to resolve the conflict until an agreement was reached on every included study. We uploaded a table to the Open Science Framework (https://osf.io/48r69/) that summarizes the details of reviewed studies.

2.4. Effect size calculation

The commonly used Fisher's z transformation was applied to transform the Pearson's r correlation coefficients in the included studies (Borenstein et al., 2009):

$$z = 0.5 \times \ln((1 + r) / (1 - r))$$

The variance of the z-transformed correlation effect size is estimated as:

$$V = 1/(n - 3)$$

3. Results

A total of 115 effect sizes were calculated across the 42 studies reported in the 37 included articles, from 11,014 participants (from K-12 to higher education). Sixteen specific self-regulated learning strategies were identified: 1, *metacognitive general* ($m = 18$); 2, *time management* ($m = 14$); 3, *effort regulation* ($m = 10$); 4, *peer learning* ($m = 2$); 5, *elaboration* ($m = 5$); 6, *rehearsal* ($m = 5$); 7, *organization* ($m = 5$); 8, *critical thinking* ($m = 8$); 9, *help seeking* ($m = 8$); 10, *cognitive general* ($m = 13$); 11, *resource management general* ($m = 2$; using available resources to support learning); 12, *self-regulation general* ($m = 22$); 13, *planning* ($m = 4$; developing a course of actions to achieve learning goals); 14, *evaluation* ($m = 2$; evaluating strengths and weaknesses of learning approaches); 15, *goal setting* ($m = 3$; setting learning goals) ($m = 3$); 16, *monitoring* ($m = 7$; monitoring progress toward goals). Among these 16 SRL strategies, Strategies 1–12 were expected from previous work using Pintrich et al.'s (1991) MSLQ, as introduced in the introduction section, and Strategies 1–9 were also found in the recent meta-analysis (Broadbent & Poon, 2015) focusing on SRL strategies in online learning environment among adults. Beyond MSLQ and Broadbent and Poon (2015), this present work also identified the other five self-regulated learning strategies (#12–16) in existing empirical work.

RQ1: Average effect of SRL strategies on academic performance in online and blended learning environments.

The overall effect size across the included studies was calculated. Conventional meta-analytical techniques require independent effect sizes, but many included studies reported multiple effect sizes from the same participants, which are dependent. Therefore, we adopted a simple random effect model via Robust Variance Estimate (RVE), a contemporary approach to performing meta-analyses on dependent effect sizes (Hedges et al., 2010).

Across the 16 SRL strategies in the included studies, the RVE model of all effect sizes indicated that the average correlation between self-regulated learning strategies and academic performance was significant, $r = .14$, with 95% confidence interval $CI_{95}[.08, .19]$ and $p < .001$. This overall effect size is considered small (<.2) based on Cohen (1988), similar to that in previous meta-analyses ($r = .13$, Broadbent & Poon, 2015; $r = .20$ across metacognitive strategies and $r = .11$ across cognitive strategies, Dent & Koenka, 2016; $r = .25$, Jansen et al., 2019; $d = .435$, Li et al., 2018). Most of the variances in observed effect sizes were estimated to reflect true differences ($I^2 = 87.23\%$; $T^2 = 0.03$). No publication bias or small-study effects were detected based on the funnel plot (see Fig. 2) and Egger's regression test ($b_{se} = 0.138$, $SE = .07$, $t(15.9) = 1.908$, $p = .075$). Lastly, the random effect was significant ($r = .12$, $p < .001$), pointing to the need to examine each of the SRL strategies separately.

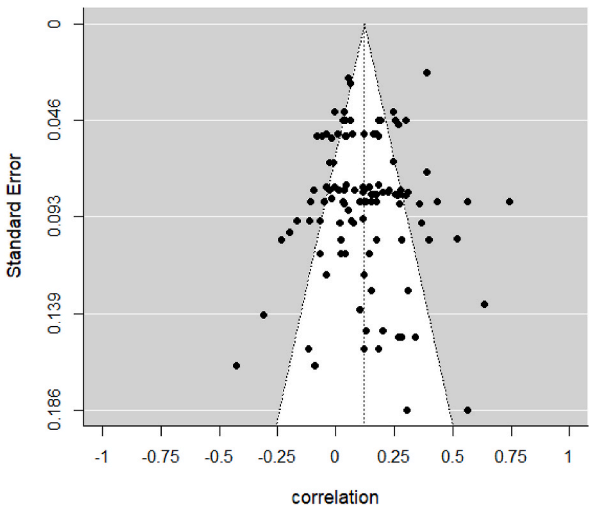

Fig. 2. Funnel plot.

RQ2: Moderator apnalyses.

RVE was also used to examine the moderation effects. Across the 16 SRL strategies, a separate mixed-effect RVE model was conducted to test the moderation effect of each of the six moderators in the relationship between SRL strategies and academic performance: article type (published journal articles vs. unpublished thesis), the type of SRL measure (questionnaires vs. behavioral coding), the type of achievement measure (existing scores in a school course or school assessment vs. other researcher-developed instruments used in research), learning context (online vs. blended), educational level (K-12 vs. undergraduate vs. graduate), and the subject type (STEM, including science, technology, engineering, and mathematics, vs. non-STEM, such as language and history) (Vo et al., 2017). The relationship between each moderator and the effect sizes was examined separately. Table 1 reports the full results. As reported below, no significant moderation effects were observed in the five moderators, suggesting that the correlations between SRL strategies and academic performance are robust to the influence of these moderators.

Article type. While the mean effect size across studies reported in peer-reviewed journal articles ($r = .15$, $p < .001$, $m = 91$) is numerically larger than that in unpublished doctoral dissertations ($r = .12$, $p = .011$, $m = 24$), this difference was insignificant ($t\,(17.40) = .41$, $p = .690$). This is consistent with the absence of publication bias indicated in the funnel plot (see Fig. 2) and Egger's regression test ($b_{se} = 0.138$, $SE = .07$, $t\,(15.9) = 1.908$, $p = .075$).

Learning context. The mean effect size across studies using blended learning ($r = .14$, $m = 22$) was the same as those used online learning ($r = .14$, $m = 93$). No significant difference was found ($t\,(7.03) = .07$, $p = .948$). This suggests that we did not observe any substantial differences in the connection between SRL strategies and academic performance across the varying degrees of online learning context.

Subject type. The effect sizes across studies focused on non-STEM subjects ($r = .14$, $m = 55$) were found comparable to that on STEM subjects ($r = .13$, $m = 26$) and that on a mixture type ($r = .14$, $m = 34$). This points to the fact that we did not detect any statistically significant differences in the relationships between SRL strategies and academic performance across different learning subjects.

Educational level. A marginally significant difference among the three educational levels was identified (HTZ (11.00) = 4.11, $p =$ .047). Most effect sizes were from studies using undergraduate samples ($m = 69$, 60%). The effect sizes were .16 ($p < .001$) and .18 ($p <$ .001) for the undergraduate and graduate subgroups, respectively. The further pairwise analyses revealed that there was no significant difference between these two groups ($p = .805$), while their effect sizes were significantly larger than that of the K-12 subgroup ($p <$ .050).

Academic measure. Nonsignificant larger effects were found between the group using existing scores such as exam scores in courses and GPA ($r = .11$, $m = 88$) and the group using measures designed by the researchers ($r = .23$, $m = 27$), $t\,(15.30) = 1.70$, $p = .110$.

SRL measure. We found that questionnaires (e.g., MSLQ) for self-regulated learning led to an effect size $r = .12$ ($m = 102$), whereas behavior coding measures resulted in a larger effect size $r = .25$ ($m = 13$). However, the difference was insignificant ($t\,(7.77) = 1.21$, p $= .261$).

Table 1
Results of moderation analyses for moderators.

Moderator	Summary effect and 95% CI						m	Test of moderation			
	r	LL	UL	t	df	p		Statistic	df	p	I^2
Article Type								$t = .41$	17.40	.690	87.42%
Journal	.15	.07	.22	4.11	28.98	<.001	91				
Thesis	.12	.03	.21	3.10	9.77	.011	24				
Learning Context								$t = .07$	7.03	.948	87.46%
Online	.14	.08	.20	4.44	34.66	<.001	93				
Blended	.14	.01	.27	2.61	4.84	.040	22				
Subject Type								HTZ = .02	22.90	.980	87.07%
STEM	.14	.05	.23	3.36	19.35	.004	55				
Non-STEM	.13	.01	.20	2.30	9.66	.038	26				
Mixed	.14	.03	.25	2.79	8.89	.022	34				
Educational level											
K-12	−.11	–	–	−1.21	3.9	–	7	HTZ = 4.11	11.00	.047	86.53%
Undergraduate	.16	.09	.23	4.80	21.2	<.001	69				
Graduate	.18	.07	.28	3.75	12.8	<.001	39				
Academic measure								$t = 1.70$	15.30	.110	80.27%
School measure	.11	.06	.16	4.27	29.56	<.001	88				
Researcher- designed measure	.23	.08	.38	3.47	8.25	.008	27				
SRL measure								$t = 1.21$	7.77	.261	87.26%
Questionnaire	.12	.07	.18	4.48	33.13	<.001	102				
Behavior coding	.25	.00	.47	2.45	5.79	.051	13				

Note. r = effect size (correlation); LL/UL = lower/upper confidence limit of 95% CI; t = t-value associated for testing statistical significance in the moderator level; df = degree of freedom after small sample correction; p = p-value associated with the t-value and df; m = number effect sizes in the respective moderator category. Statistic (test of moderation): t value for single parameter tests or Hotelling-T approximated (HTZ) test statistic for multiple parameter tests. The value of I^2 reflects the proportion of true variance in the total observed variance of effect sizes explained by the respective moderator. Significance tests for the summary effects are not interpretable when $df < 4$, so the p-value and confidence interval values were not reported for the K-12 subgroup in the *educational level* analysis.

RQ3: Average effect for each of 16 specific SRL strategies.

The RVE approach was originally used to estimate the effect sizes of SRL strategies on academic performance. As we have two different categories of SRL strategies, we calculated the average effect for the sixteen specific strategies and for the four main types of SRL strategies. However, the sample sizes for each specific SRL strategy were small when divided into sixteen categories; in this case, the degrees of freedom for several groups fell below 4, making the significance tests statistically uninterpretable (Tipton & Pustejovsky, 2015). The meta-analyses using random effect models were, therefore, conducted to calculate the effect sizes of these sixteen specific SRL strategies. Six out of thirty-seven articles reported more than one coefficient for a single SRL strategy, and we used the composite scores by averaging the coefficients from the same SRL strategy from one study.

Table 2 shows the results of random effect subgroup analysis over the sixteen specific categories of SRL strategies. The observed effect sizes, in accordance with Cohen's (1998) classification, generally exhibited small magnitudes, ranging from .00 to .18. Strategies that significantly correlated with academic performance were *effort regulation* ($r = .18$, $p < .001$), *metacognitive strategies* ($r = .17$, $p < .001$), *time management* ($r = .17$, $p < .01$), *organization* ($r = .16$, $p < .001$), *general SRL strategies* ($r = .15$, $p < .01$), and *rehearsal* ($r = .08$, $p < .01$).

RQ4: Average effect for four major types of SRL strategies.

The sixteen specific SRL strategies were grouped into four major categories according to the definitions in Pintrich et al.'s (1991) MSLQ as well as our included studies: five types of cognitive strategies (*rehearsal*, *elaboration*, *organization*, *critical thinking*, and *general cognitive strategies*), five types of metacognitive strategies (*planning, goal setting, monitoring, evaluation,* and *general metacognitive strategies*), five types of resource management strategies (i.e., *time management*, *effort regulation*, *peer learning*, *help seeking,* and *general resource management*), and one type of *general SRL strategies* (unspecified SRL strategy). Table 3 shows the results of random effect subgroup analysis over the four major types of SRL strategies; the effect sizes ranged from .09 to .15, which fell into small effect sizes according to Cohen (1998). *Cognitive strategies, metacognitive strategies*, and *resource management strategies* significantly correlated with academic achievement, with correlation values of .09 ($p < .05$), .13 ($p < .01$), and .14 ($p < .001$), respectively. Moreover, *SRL general strategies* were significantly correlated with academic achievements ($r = .15$, $p < .01$).

4. Discussion

The present meta-analysis sought to clarify the relationship between students' use of SRL strategies and their academic performance in online and blended learning environments. We identified forty-two studies from thirty-seven articles and analyzed the included 115 effect sizes from 11,014 participants in K-12 and higher education. Our key findings include: (1) the average correlation between SRL strategies and academic performance was statistically significant, with a value of $r = .14$; (2) no significant moderation effects were detected for article type, learning context, subject type, SRL measure, or measure of academic performance in this relationship; (3) rehearsal, organization, metacognitive strategy, time management, effort regulation, and general SRL strategies are effective strategies associated with academic performance; (4) on theoretical level, all four types of SRL strategies exhibited significant correlations with academic performance.

Our meta-analysis supports a significant yet relatively small overall mean correlation ($r = .14$) between SRL strategies and academic performance, encompassing students of varying age groups in online and blended learning environments. Notably, this effect size was numerically smaller than previous meta-analysis conducted in traditional in-person learning environments (e.g., $r = .20$ for the overall correlations of metacognitive strategies and academic achievement among children and adolescents, respectively, Dent & Koenka, 2016; $r = .20$ in higher education, Jansen et al., 2019). This finding is consistent with the previous meta-analysis that focused on the online learning environment among adults ($r = .13$, Broadbent & Poon, 2015). Importantly, we extend this conclusion to encompass a broader age range of students (from children to adults), and encompass blended learning environments, which integrate

Table 2
Results of random effect subgroup analysis over SRL strategies (16 categories).

SRL strategies	*m*	*r*	95% C.I.	*z*-value	*p*-value
Rehearsal	5	.08**	[.03, .14]	2.98	.003
Elaboration	5	.09	[-.04, .22]	1.38	.169
Organization	5	.16***	[.11, .21]	5.74	<.001
Critical thinking	8	.00	[-.05, .04]	−.19	.848
Cognitive general	13	.14	[-.00, .27]	1.95	.051
Metacognition	18	.17***	[.08, .25]	3.57	<.001
Time management	14	.17***	[.07, .26]	3.47	<.001
Effort regulation	10	.18***	[.10, .25]	4.34	<.001
Peer learning	2	.03	[-.05, .11]	.74	.462
Help seeking	8	.08	[-.01, .16]	1.75	.081
Self-regulation general	22	.15**	[.05, .25]	2.82	.005
Planning	4	.13	[-.07, .31]	1.27	.204
Evaluation	2	.02	[-.15, .18]	.19	.851
Goal setting	3	.04	[-.15, .23]	.43	.667
Monitoring	7	.07	[-.24, .36]	.41	.682
Resource management	2	.15	[-.08, .35]	1.29	.196

Note. m = number effect sizes in the respective moderator category; r = effect size (correlation).

Table 3
Results of random effect subgroup analysis over SRL strategies (4 levels).

SRL strategies	*m*	*r*	95% C.I.	*t*-value	*p*-value
Cognitive	37	.09*	[.04, .13]	2.30	.036
Metacognition	33	.13**	[.06, .20]	3.26	.004
Resource management	36	.14***	[.09, .19]	4.68	<.001
Self-regulation general	22	.15**	[.05, .25]	2.84	.005

Note. m = number effect sizes in the respective moderator category; *r* = effect size (correlation).

online learning components into the educational process.

4.1. Moderation effects

Our study did not find article type, SRL measure, achievement measure, learning context, and subject type to significantly moderate effect sizes. This is consistent with previous studies regarding article type (Xu et al., 2022), SRL measure (Jansen et al., 2019), and subject type (Dignath et al., 2008; Jansen et al., 2019; Zheng, 2016). However, some previous studies found the opposite, which suggested a significant moderator effect on achievement measure (Jansen et al., 2019), learning context (Xu et al., 2022), and subject type (Li et al., 2018). Our findings indicate that the impact of SRL strategies may not vary significantly across various measures, learning contexts, and disciplines. However, the broad categorizations used for learning contexts and disciplines may obscure more nuanced effects. Future research should explore whether specific characteristics of these two contextual factors, such as the availability of technical support, influence the effectiveness of SRL strategies. Educational level demonstrated a marginally significant moderation effect, with a lower effect size for K-12 students than for undergraduates and graduates. This finding indicates that the impact of enhanced SRL strategies on academic performance seems to be more pronounced among adults than among K-12 students. This difference might be attributed to the development of SRL over time and the change in the learning environment across grades. Students' ability to regulate their learning has been found to improve throughout childhood and adolescence (Paris & Newman, 1990). Moreover, students' academic success in college often relies on SRL skills more heavily, as they typically receive less guidance from teachers and face more distractions than in K-12 years (Brady et al., 2021). However, it is important to note that while SRL skills may be less developed in K-12 students than in adults, their academic performance can still benefit from improved SRL abilities. This is supported by a recent meta-analysis that identified a stronger effect size of SRL interventions on academic learning for primary school students than for older students (Xu et al., 2022). Therefore, we call for increased attention to and support for SRL in young students.

For the magnitude of effect sizes, our study is consistent with previous studies regarding published journals ($d = .63$) being higher than unpublished ($d = .49$) (Xu et al., 2022). SRL measure indicates that the mean effect size for behavior during the task ($r = .32$) is larger than that from the questionnaire ($r = .14$, Dent & Koenka, 2016), while inconsistent regarding STEM subjects ($d = .82$, Dignath et., 2008; $r = .458$, Zheng, 2016; $d = .88$, Xu et al., 2022) being larger than non-STEM subjects ($d = .55$, Dignath et., 2008; $r = .418$, Zheng, 2016; $d = .40$, Xu et al., 2022). The different findings might be due to two main factors: (1) the validity of performance measures for STEM and non-STEM subjects and (2) the influence of learning contexts (face-to-face or online and blended learning) on educational effectiveness.

4.2. The average effect size for specific and four categories of SRL strategies

We formed comparisons on the relationships between different SRL strategies and academic performance at two different levels: sixteen specific SRL strategies and four categories of SRL strategies (based on Pintrich et al., 2004). Our findings on these two levels of analysis appear to be consistent: (1) *metacognitive strategies* were a significant predictor of academic achievement across two levels of analyses; (2) *resource management* and two of its four specific strategies (*effort regulation* and *time management*) significantly predicted academic achievement, except for peer learning; (3) *cognitive strategies* were not significant as a grouped category, although two of its five specific strategies (*rehearsal* and *organization*) were significant predictors of academic performance; (4) *general SRL strategies* were significant as at both two analysis levels; (5) no significant moderating effects were found except for educational level.

Our study revealed that rehearsal, organization, metacognitive strategy, time management, effort regulation, and general SRL strategies are effective strategies associated with academic performance. It is consistent with previous studies regarding rehearsal ($r = .08$, Dent & Koenka, 2016), organization ($r = .08$, Dent & Koenka, 2016), metacognitive strategy ($r = .06$, Dent & Koenka, 2016; $r = .18$; Richardson et al., 2012; $d = .388$, Li et al., 2008), time management ($r = .14$, Broadbent & Poon, 2015), effort regulation ($r = .11$, Broadbent & Poon, 2015). We did not find any significance in some specific strategies (elaboration, critical thinking, help-seeking, planning, evaluation, and goal setting). At the same time, some other previous meta-analyses reported significant findings related to those strategies (Broadbent & Poon, 2015; Dent & Koenka, 2016; Li et al., 2018; Richardson et al., 2012). The discrepancies between our effect sizes and those of previous studies may be attributed to the complexity of current SRL research and its interdisciplinary nature. SRL encompasses multiple dimensions, including cognitive, metacognitive, and resource management, across diverse learning contexts, such as online, offline, and blended learning environments. Even within online settings, distinctions exist between asynchronous and synchronous formats. However, most previous reviews did not differentiate the learning context into more specific categories. The interdisciplinary nature of SRL research lies in its integration of fields like educational psychology, computer science, and learning analytics (Panadero, 2017; Winne & Hadwin, 2013). The rise of digital tools and learning platforms, such as learning management systems (LMS), has enabled the capture of behavioral data, often referred to as digital traces. This advancement provided new approaches to the analysis and interpretation of SRL processes (Winne & Hadwin, 2013). However, our included studies primarily utilized traditional questionnaires, think-aloud protocols, and content analysis. This limited range of SRL measures suggests that current research does not fully capture the complexity of SRL mechanisms, leading to variations in observed effect sizes.

Each previous meta-analysis study had different SRL models, SRL definitions, and SRL categorizations. Thus, the same terms for SRL strategies may have different definitions across previous meta-analyses. For example, the definition of time management in Richardson et al. (2012) is much broader than in Sizmann and Ely (2021). For Richardson et al. (2012), effort regulation means persistence and effort, whereas for Sitzmann and Ely (2011), it is simply time devoted to a task. In addition, our study includes unspecified SRL strategies as part of general SRL strategies. In other studies, however, this is categorized as a specific strategy.

Moreover, our study has similarities with previous studies regarding the effect sizes of cognitive strategy (e.g., r = .11, Dent & Koenka, 2016), metacognitive strategy (r = .20, Dent & Koenka, 2016; d = .19, Xu et al., 2022), and resource management strategy (d = .46, Xu et al., 2022). The difference in the magnitude of the effect size may be because of the different definitions and categorizations. For example, cognitive strategies in our study included rehearsal, elaboration, organization, critical thinking, and general cognitive strategies. In contrast, Dent and Koenka (2016) included three strategies that are the same as ours, as well as nine other different strategies, such as deep strategies and surface strategies. For resource management strategies, our study included time management, effort regulation, peer learning, help-seeking, and general resource management. However, only two of the same strategies are shared by other studies (time management and help-seeking). Three other strategies are different (environment structuring, environment structuring, time management, persistence) in Jansen et al. (2019). As such, it is possible that the different definitions and categorizations resulted in different magnitudes of effect size for each type of strategy.

Another possible reason to explain the effect size differences in specific strategies might be due to the different learning contexts (face-to-face or online and blended learning). Our study demonstrated the significant effect of metacognitive strategy in face-to-face (Li et al., 2018; Richardson et al., 2012) and online and blended learning contexts. Future research should focus on the importance of metacognitive strategy in different contexts. For time management, researchers found small but insignificant effects in traditional learning contexts (Dent & Koenka, 2016; Sitzmann & Ely, 2011), while small and significant effects were found in online settings (Broadbent & Koenka, 2016). Our findings indicate that time management has a small and significant effect on academic performance, which validates the effectiveness of time management strategies for online and blended environments.

Despite the complexity of the SRL strategies, we found two general patterns across studies. First, superficial strategies generated a small effect size. For instance, rehearsal in our study generated a small but significant correlation (r = .08), which aligns with the findings from previous studies (r = .08, Dent & Koenka, 2016; r = .01, Richardson et al., 2012). Cognitive strategies like rehearsal are designed to assist learners in acquiring information at a superficial level through the retention of knowledge. The rehearsal strategy is considered a shallow and superficial strategy that supports the surface processing of information, which needs further in-depth processing to facilitate meaningful learning (Garcia & Pintrich, 1994; Pintrich, 2000). Thus, shallow processing strategies are expected to have a weak association with achievement. As Garcia and Pintrich (1994) pointed out, deep strategies like organization (e.g., using flowcharts or outlines) promote better performance. In this regard, it was supported by our finding that organization strategy (r = .16) had a larger effect size than rehearsal (r = .08).

Moreover, higher-order strategies like metacognitive strategies seem to be essential. Our study reported a significant prediction for metacognitive strategy (r = .17), which was consistent with previous studies (r = .06, Dent & Koenka, 2016; r = .18, Richardson et al., 2012; d = .39, Li et al., 2008). Metacognition is a comprehensive concept that encompasses different aspects of self-regulation. Despite the various definitions of metacognition, the critical role of metacognition has been well recognized. According to Hattie et al. (1996), the findings suggested that interventions were most successful when implemented within a specific context and promoted active student engagement and metacognitive awareness. Several included studies have demonstrated the vital role of metacognitive strategy. For example, Cho and Shen (2013) identified an effect size of .15 of SRL on student achievement in an online course. In Dunnigan (2018), SRL was investigated in a community college program in a required online math course, and the correlation between metacognitive strategy (r = .20) and academic performance was reported. In Pickett (2018), SRL was explored in online adult education among 137 adult learners, and a correlation of .034 was reported for the metacognitive strategy.

Furthermore, compared to the individual type of strategy alone, our findings suggest that combined SRL strategies (e.g., SRL general, metacognitive) tend to have larger effect sizes.

Previous studies have indicated better effects for combined types of strategies (Dignath & Büttner, 2008; Dignath et al., 2008; Jansen et al., 2019; Xu et al., 2022). A stronger relationship between SRL and achievement was found when metacognitive and resource management were combined (Jansen et al., 2019). The highest effects were found for combined instruction with different types of strategies (Dignath et al., 2008; Dignath & Büttner, 2008). The effect of using a mixed SRL strategy was higher than another individual strategy (Xu et al., 2022). Therefore, the role of combining different types of strategies can be explored in future research.

5. Conclusion

To sum up, as for the 16 specific strategies, rehearsal, organization, metacognitive strategy, time management, effort regulation, and general SRL strategies were found to be effective strategies associated with academic performance. Mixed types of SRL strategies were also recommended. Our study confirmed that superficial strategies usually generate small effect sizes and that metacognitive strategy is essential in both online and blended learning environments. The reasons for different effect sizes might be due to different definitions and categorizations and different learning contexts. All four SRL strategies categorized on the theoretical level showed significant correlations with academic performance. Our study did not find significant moderation effects on the relationships between

SRL strategies and academic performance; the only exception was educational level, with a lower effect size for K-12 students than for adults. Our study validated that SRL strategies are suitable not only for face-to-face traditional classroom settings but also for online and blended environments.

One should be mindful of the traditional measures being used by many of the studies in this review. While these measures are suitable for the traditional face-to-face classroom, they may not translate to how students learn in the online environment. For instance, 19 of the 37 included studies used the MSLQ. As noted in Broadbent and Poon (2015), the validation of the MSLQ had been within traditional face-to-face settings; it is unclear to what extent this measure may capture the construct of learner self-regulation accurately in a different learning context, such as the online environment. Also, there is a lack of validation information for researcher-designed assessments measuring academic performance. Only two studies detailed the validation procedures for their achievement measures. Future studies should include comprehensive validation information when using researcher-designed assessments. Another notable limitation of this study is the exclusion of motivational and emotional aspects of SRL from the scope. While this decision enhanced comparability with previous studies, it may limit a more comprehensive understanding of SRL strategies. Emotional regulation, in particular, plays a critical role in activating the learning pathway and facilitating the effective use of learning strategies (Boekaerts, 2011; Pekrun, 2017). Further research may consider specifically examining the effects of motivational and affective components of SRL on students' academic achievement in online and blended learning contexts. Moreover, an important consideration in interpreting the results is the influence of sample size on the observed effect sizes for certain SRL strategies. Strategies such as planning and evaluation included in this meta-analysis were not extensively examined in prior studies and had smaller sample sizes. This may potentially reduce the statistical power to detect significant effects. Future studies are encouraged to focus on these less-studied strategies to better understand their impact on academic performance. Lastly, the majority of the studies we included examined SRL and academic performance in higher educational settings. Although we found a marginally significant moderation effect of educational level, the result should be interpreted with caution due to the small sample size of effect sizes from K-12. This also highlights the need for further attention to younger students, especially the development of instruments that match their developmental stage.

CRediT authorship contribution statement

Yingying Zhao: Writing – review & editing, Writing – original draft, Project administration, Formal analysis, Data curation, Conceptualization. **Yixun Li:** Writing – review & editing, Writing – original draft, Supervision, Methodology. **Shuai Ma:** Writing – review & editing, Writing – original draft. **Zhihong Xu:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation. **Bingsheng Zhang:** Writing – review & editing, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could be constructed as a conflict of interest.

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Data availability

Data will be made available on request.

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